

Bill Organizer Project Proposal

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A) Your selected data type

Our selected datatype is a receipt/bill organizer. An app that will help households to keep track of their bills such as water, electricity, Wi-Fi, etc.

B) The operations you will support

1. Add bill
2. Edit bill
3. Delete bill
4. View bills
5. Filter bills
6. Sort bills
7. Search
8. Save/Load (binary)
9. Reports & summaries
10. Export CSV
11. Quit

C) The user interface design (be creative here, but only use the C standard library)

```
Printf("-$$----Bill Organizer----$$-\n");  
  
Printf("|  
|  
| 1) Add bill  
| 2) Edit bill  
| 3) Delete bill  
| 4) View & filter bill  
| 5) Sort bill  
| 6) Reports & summaries |-\n");
```

```

Printf("| 7) Save (binary)      |\n");
Printf("| 8) Load (binary)     |\n");
Printf("| 9) Export CSV           |\n");
Printf("| 0) Quit                  |\n");
Printf("|                          |\n");
Printf("-----\n");

```

```

Printf("Choice: _\n");

```

- Add bill
 - Prompts in order
 - Category --> Store/Provider --> Total --> Payment Item --> Payment Method --> Date--> Billing Period --> Tax relevance --> Notes --> Confirm
- View & Filter
 - Filter panel
 - Last paid
 - Category, Provider
 - Tax tag (Yes, No)
 - Min/Max Total
- Reports & summaries
 - “By Month” table (Count, Total, Average)
 - “By Category” table (Count, Total, Min/Max/Avg)
- Accessibility & UX
 - Show how many records match the filter
 - Two decimal places
 - Clear status badges
 - Confirmations before delete and overwrite

D) The structure of the source code

1. billLib.h

- i. This file will contain the prototypes of all the functions that implement the functionalities described in the project document for the Bill Organizer.
 - ii. It will contain the declaration of the main data structure used to store all the bills in memory
- 2. billLib.c
 - i. This file will contain the implementations of all the functions whose prototypes are declared in billLib.h, specifically focused on the Bill Organizer logic:
- 3. main.c
 - i. This file will contain the main function, which will:
 - a. Render the main menu that the user will use to interact with the Bill Organizer
 - b. Read the user's selection and, based on that choice, call the appropriate functions defined in billLib.h/c to perform the requested operations
 - c. Loop the menu continuously until the user chooses the exit option

E) The distribution of the development among the individual team members

Alex:

- i. Add bill
- ii. Edit bill
- iii. Delete bill

Ben:

- i. View & filter
- ii. Sort
- iii. Save / Load

Fabian:

- i. Reports & summaries
- ii. Export CSV
- iii. GUI